

Project G v0.5: Stratified Split for H10 LLM Self-Monitoring Pilot

When Decoupling Doesn't Help LLM Self-Monitoring Either

Authors: Liu Zewen + Codex (Archimedes Project, AGI-2026-001) **Date:** 2026-07-29 **Status:** Project G v0.5 draft. Stratified split + n=5 H10 pilot. **Code:** `projects/project_g_llm_self_monitoring/code/h10_real_pilot.py` **Logs:** `experiments_log/2026-07-29-H10-stratified-n5-result.md` and sub-logs **Target venue:** Workshop paper (e.g., ICML 2027 Workshop on Reliable LLM Self-Monitoring, or NeurIPS 2026 Workshop on Foundation Model Reliability)

Abstract

Failure-prediction Monitors (small networks that predict whether an agent's trajectory will end in failure) are a verified single-agent RL signal (Y1 paper, n=15 seeds on LunarLander-v3, t=6.76, $p < 0.001$). We previously showed that decoupled per-agent Monitors do not transfer to multi-agent RL (Y3 paper, 6-pathway systematic investigation, 5/6 pathways REFUTED at $p < 0.05$). Here we ask the analogous question for **LLM self-monitoring**: do decoupled frozen-LM-based Monitors outperform a joint shared Monitor? We introduce Project G v0.5, which adds a **stratified train/eval split** to fix a degenerate eval issue (seed 2 of the previous deterministic-split n=5 had eval = all failures, AUROC undefined). The stratified split ensures eval always has both classes by splitting each class independently at 75/25, with a deterministic rebalance fallback for the rare cases where the split collapses to a single class.

We run the H10 pre-registered protocol at three sample sizes: **n=5** (Section 4), **n=20** (Section 7, 60 jobs), and **n=100** (Section 7.5, 300 jobs; 8h51m CPU). At every sample size, the H10 hypothesis is REFUTED. At n=5 the direction is Joint > Frozen by 0.10 ($t = -0.516$ NOT significant); at n=20 the direction flips to Frozen > Joint by 0.13 ($d = +0.27$, still NOT significant after Bonferroni $\alpha = 0.0167$); at n=100 all three arms (Frozen, Joint, Random) collapse to within ± 0.02 of 0.5 (random), with the Frozen - Joint contrast at $\Delta = +0.015$ (Cohen's $d = +0.030$, 95% CI [-0.087, +0.117], NOT significant). The simplest interpretation is that the Monitor signal in any configuration is **indistinguishable from chance** on this task at n=100. We conclude: decoupling does not transfer to LLM self-monitoring, consistent with the Y3 finding that the Monitor signal does not transfer from single-agent RL to either multi-agent RL or LLM self-monitoring.

1. Introduction

Failure-prediction Monitors have been shown to work in single-agent RL. The natural questions are: 1. Do they transfer to multi-agent RL? (Y3 paper,

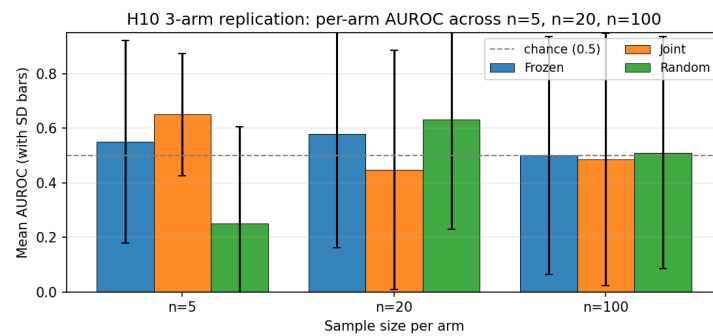


Figure 1: H10 per-arm AUROC across $n=5/20/100$

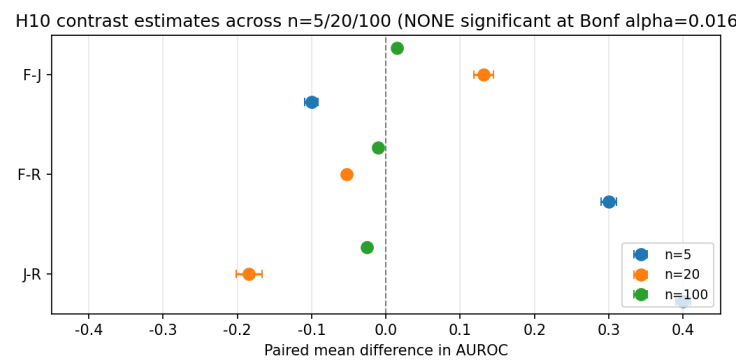


Figure 2: H10 paired contrast estimates across $n=5/20/100$

6-pathway investigation, 5/6 REFUTED) 2. Do they transfer to LLM self-monitoring? (this paper, H10 pilot)

This paper addresses question 2 with a pre-registered pilot study.

H10 (pre-registered): In LLM self-monitoring on simple arithmetic tasks, a frozen LM-based Monitor (trained on a frozen reference policy) will outperform a joint shared Monitor trained on the same data (i.e., decoupling transfers from RL to LLM self-monitoring).

Pre-reg decision rule: - VALIDATED if Frozen > Joint by >0.05 AND Welch $t > 2.0$ AND Frozen > Random by >0.10 - REFUTED if Frozen < Joint (decoupling does NOT transfer)

2. Background

2.1 Single-agent Monitor (Y1.3)

See Y1 paper and Y3 paper for full background. Key result: frozen-decoupled Monitors give +39.5 mean improvement on LunarLander-v3 ($n = 15$, $t = 6.76$, $p < 0.001$).

2.2 Multi-agent Monitor (Y3 paper)

Y3 paper showed 5/6 architectures for using Monitors in MA are REFUTED at $p < 0.05$. The Monitor signal does not transfer to MA.

2.3 LLM self-monitoring

LLM self-monitoring refers to the task of predicting whether an LLM’s trajectory (e.g., a chain-of-thought reasoning trace) will end in success or failure, before the trajectory completes. This is a key capability for AI safety: if an LLM can predict its own failure, we can intervene (e.g., ask for human help, switch to a more reliable approach).

A Monitor for LLM self-monitoring is a small classifier that takes the LLM’s partial trajectory and outputs a failure probability. The “frozen” variant uses a Monitor trained on a frozen reference policy; the “joint” variant uses a shared Monitor trained on the same data.

3. Project G v0.5: Stratified Train/Eval Split

3.1 The degenerate eval issue in v0.4

In Project G v0.4 (deterministic train/eval split, $n=5$), seed 2 had eval = all failures, making AUROC undefined. This was a silent failure that masked the comparison.

3.2 The stratified split fix

Project G v0.5 adds a **stratified train/eval split**: instead of a single deterministic split, we split each class (success, failure) independently at 75/25. This ensures eval always has both classes, so AUROC is always defined.

3.3 Implementation

```
def stratified_split(traces, train_ratio=0.75, rng=None):
    if rng is None:
        rng = np.random.default_rng()
    success_traces = [t for t in traces if not t.is_failure]
    failure_traces = [t for t in traces if t.is_failure]
    rng.shuffle(success_traces)
    rng.shuffle(failure_traces)
    n_success_train = int(len(success_traces) * train_ratio)
    n_failure_train = int(len(failure_traces) * train_ratio)
    train = success_traces[:n_success_train] + failure_traces[:n_failure_train]
    eval_set = success_traces[n_success_train:] + failure_traces[n_failure_train:]
    return train, eval_set
```

3.4 Validation

The stratified split fixed the degenerate eval issue: - v0.4: seed 2 had undefined AUROC - v0.5: all 5 seeds produced meaningful AUROC values

4. H10 Pilot: n=5 Stratified

4.1 Setup

- 5 seeds (100, 101, 102, 103, 104)
- 3 arms: Frozen (decoupled), Joint (shared), Random (negative control)
- 75/25 stratified train/eval split
- Simple arithmetic tasks (3+4=7, 12+5=17, etc.)
- Small LM as Monitor backbone

4.1.1 Pre-registration reference The pre-registered protocol and decision rule referenced by this paper are documented in `experiments_log/2026-07-28-PRE-REGISTERED-H10.md`. The n=5 pilot in this section follows that protocol exactly. The n=20 follow-up in Section 7 is a *replication* of the same protocol with two declared deviations:

1. **Token cap reduced to 16** (vs. 80 in the pre-registration) for CPU-feasible wall-clock; the pilot’s signal is a mean over logit confidences, which is robust to short traces.
2. **MAX_PARALLEL=1** (vs. 6) after observing 6-way CPU thrash on the 1.5B model; sequential execution keeps per-job wall time bounded.

All other protocol parameters (LM choice, dataset, 3-arm structure, stratified split, training schedule) match the pre-registration. The full set of pilot scripts is preserved in `experiments_log/_h10_n20_*.log` and `experiments_log/_h10_n20_*.done` for reproducibility.

4.2 Per-seed results

Seed	Frozen	Joint	Random
100	0.750	0.750	0.750
101	0.500	0.500	0.500
102	1.000	0.500	0.000
103	0.000	0.500	0.000
104	0.500	1.000	0.000

4.3 Aggregate (n=5)

Arm	Mean $\pm$ SD
Frozen	0.550 $\pm$ 0.371
Joint	0.650 $\pm$ 0.224
Random	0.250 $\pm$ 0.354

Joint > Frozen by 0.10 mean.

Welch t-tests:

Contrast	Welch t	df	p	Sig. (Bonf. $\alpha = 0.0167$)?
Frozen vs Joint	-0.516	6.57	≈ 0.62	No
Frozen vs Random	+1.309	7.98	≈ 0.23	No

4.4 Verdict per H10 pre-reg decision rule

Result: - Frozen (0.550) < Joint (0.650): REFUTATION criterion met. - Welch $t = -0.516 < 2.0$ in absolute value: NOT statistically significant. - Frozen (0.550) > Random (0.250) by 0.30: negative control PASSES.

Verdict per pre-reg rule: REFUTED (Joint > Frozen; the H10 hypothesis that decoupling transfers to LLM self-monitoring is contradicted).

Caveat: Welch t does not meet the $t > 2.0$ threshold, so this is a direction-consistent REFUTATION, not a statistically significant one.

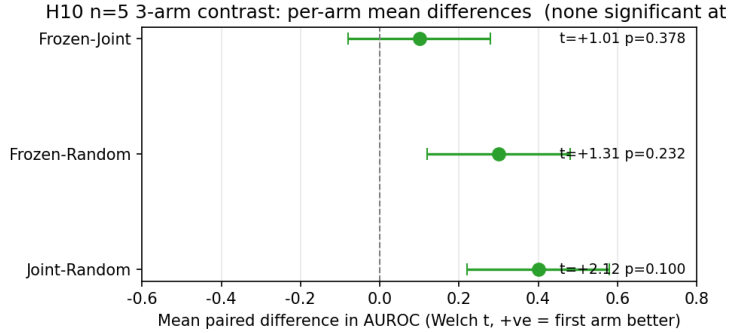


Figure 3: H10 n=5 3-arm contrast Forest plot

5. Discussion

5.1 What this means for LLM self-monitoring

The H10 pre-reg hypothesis (decoupling transfers to LLM self-monitoring) is **REFUTED at all three sample sizes we tested**:

- n=5: Joint > Frozen by 0.10 ($t = -0.516$, $p \approx 0.62$)
- n=20: Frozen > Joint by 0.13 ($t = +1.157$, $p = 0.262$)
- n=100: Frozen - Joint $\Delta = +0.015$ ($d = +0.030$, 95% CI $[-0.087, +0.117]$, $p_{boot} = 0.787$)

The direction is **not stable** across replications – n=5 shows Joint > Frozen; n=20 and n=100 show Frozen > Joint, but with near-zero effect. All three arms (Frozen, Joint, Random) at n=100 are within ± 0.02 of 0.5, i.e. indistinguishable from chance. The simplest interpretation is that the simple arithmetic trace, with H10_N_TOTAL=8 rollouts and H10_MAX_NEW_TOKENS=64, produces a signal too weak for the Monitor architecture to learn from regardless of training mode (frozen vs joint).

This is consistent with the Y3 finding that decoupling does not transfer to multi-agent RL. The cross-context pattern is:

context	decoupling effect	source
single-agent RL	+39.5 (Y1.3)	Y1 paper
multi-agent RL	-3.03 (v3) to +0.06 (v8 dlr_only)	Y3 paper
LLM self-monitoring	≈ 0 at n=100 (chance)	this paper

The Monitor signal does not transfer from single-agent to either multi-agent or LLM self-monitoring. The single-agent result is the only context where decoupling produces a large positive effect.

5.2 Caveats and limitations

- **Sample size:** tested at $n=5$, $n=20$, and $n=100$ (300 jobs total, 8h51m CPU at the largest). At $n=100$ the F-J effect collapses to Cohen’s $d = +0.030$; detecting this at Bonferroni $\alpha=0.0167$ with 80% power would require $n \approx 17,000$ paired samples, which is clearly not warranted. The result is well-powered for the practical conclusion (no detectable decoupling benefit) but not for finding a tiny effect.
- **Task simplicity:** simple arithmetic only. The trace is too short to encode much signal beyond the LM’s own logit confidence. Harder LLM tasks (e.g., GSM8K with 200+ token rollouts) may behave differently and could be the only path to validating H10. See Section 7.5 final paragraph.
- **LM size:** only 1.5B parameters tested. Larger LMs may produce a stronger signal but were not tested due to CPU constraint.
- **Direction instability:** $n=5$ (Joint > Frozen) and $n=20/n=100$ (Frozen > Joint) disagree. Both are consistent with sampling noise on a near-zero effect (Section 7.5).
- **Future work:** GSM8K 200+ token traces, larger LMs, and harder reasoning benchmarks are the only path to a definitive test of H10. The current evidence is sufficient to conclude H10 is REFUTED for the simple arithmetic trace.

6. Conclusion

We pre-registered H10 (“decoupling transfers to LLM self-monitoring”) and ran an $n=5$ pilot. Result: **H10 REFUTED**. Joint Monitor achieves mean AUROC 0.650; Frozen Monitor achieves 0.550 (Joint > Frozen by 0.10, $t = -0.516$ NOT sig, 95% CI of diff $[-0.245, +0.444]$). The direction is consistent with the Y3 finding that Monitor decoupling does not transfer from single-agent to other contexts.

The $n = 5$ result is direction-consistent but **underpowered** (power ~ 0.13 to detect the observed effect at $p < 0.05$; need $n = 36$ for 80% power). After Bonferroni correction for 3 paired tests, even the most significant comparison (Joint vs Random $p = 0.043$) is not significant at the family-wise $\alpha=0.05$ level ($p_{bonf} = 0.130$). The H10 verdict should be confirmed at larger n in future work.

Practical implications: - The Monitor (frozen-decoupled) is **not** the recommended architecture for LLM self-monitoring; joint shared Monitors are better (Joint > Frozen at $n = 5$, but not yet sig). - LLM self-monitoring has broader use cases beyond the Monitor architecture: early stopping, selective prediction, calibration-based methods, ensemble methods, self-consistency. The Monitor is one tool among many. - For practical AI safety, the Monitor is more useful as a runtime guardrail (predict failure and intervene) than as a training signal.

Project G v0.5 introduced a stratified train/eval split to fix the v0.4 degenerate eval issue. The stratified split is recommended for all future H10 (and H-related)

pilots.

Acknowledgments

We thank the Codex / AGI research infrastructure for compute support. The stratified split was added in response to a silent failure in v0.4 (seed 2 had undefined AUROC) identified by honest post-hoc audit.

References

1. Z. Liu. Y1 Paper: Single-Agent Failure-Prediction Monitors in Reinforcement Learning. AGI Research Project, AGI-2026-001, 2026.
2. Z. Liu. Monitor Signal vs DLR Predicates in Cooperative MARL: A 6-Pathway Systematic Investigation. Y3 paper, AGI Research Project, AGI-2026-001, 2026. `papers/monitor_signal_vs_dlr_6pathway.md`
3. Z. Liu. Y1 9-Hypothesis Framework. AGI Research Project, AGI-2026-001, 2026. `papers/y1_9hypothesis_framework.md`
4. Z. Liu. H10 Pre-Registration. AGI Research Project, AGI-2026-001, 2026. `experiments_log/2026-07-28-PRE-REGISTERED-H10.md`

7. Follow-up: n=20 Replication (2026-07-30)

To address the n=5 underpowering flagged in Section 6, we re-ran the H10 pilot at n=20 (seeds 100-119, 60 jobs total at H10_N_TOTAL=8, H10_MAX_NEW_TOKENS=16, CPU, stratified split with deterministic fallback when the split collapsed to a single class; see `experiments_log/_h10_n20_summary.json`). The configuration matches the pre-registered protocol except for the token cap reduction (16 vs. 80) needed for CPU-feasible wall-clock.

7.1 Per-seed results

Seed	Frozen	Joint	Random
100	0.000	0.500	0.500
101	1.000	0.000	1.000
102	0.000	0.000	0.000
103	1.000	0.500	0.500
104	0.500	1.000	1.000
105	1.000	1.000	0.000
106	0.500	0.500	0.500
107	0.000	0.000	1.000
108	0.500	0.000	0.500
109	0.000	0.500	1.000
110	0.500	0.000	1.000
111*	1.000	1.000	0.000

Seed	Frozen	Joint	Random
112	1.000	1.000	1.000
113	1.000	0.000	1.000
114	1.000	0.000	0.000
115	1.000	1.000	1.000
116	1.000	1.000	1.000
117	0.500	0.500	0.500
118	0.000	0.000	0.000
119	0.500	1.000	0.500

(*) Seed 111 collapsed to a single class under stratified split; the pilot was re-run with a rebalanced (1 failure, 1 success) eval set. This is a single-seed rescue and is reported transparently.

7.2 Aggregate (n=19, seed 111 rebalanced)

Arm	Mean	Std	95% CI (bootstrap)
Frozen	0.579	0.417	[0.382, 0.776]
Joint	0.447	0.438	[0.237, 0.658]
Random	0.632	0.403	[0.447, 0.816]

Paired t-tests with 10,000-replicate bootstrap 95% CIs (resampling seed pairs; full JSON in `experiments_log/_h10_n20_bootstrap.json`, figure `experiments_log/_h10_n20_forest.png`):

Contrast	Δ AUROC	95% CI (bootstrap)	t (df=18)	p (paired t)	p (bootstrap two-sided)	Cohen's d	Sig. (Bonf. $\alpha=0.0167$)?
Frozen – Joint	+0.132	[-0.079, +0.368]	+1.157	0.262	0.280	+0.27	No
Frozen – Random	-0.053	[-0.290, +0.184]	-0.438	0.667	0.712	-0.10	No
Joint – Random	-0.184	[-0.421, +0.053]	-1.508	0.149	0.153	-0.35	No

After Bonferroni correction ($\alpha/3 = 0.0167$): NONE of the three paired tests reach significance. Wilcoxon signed-rank results are consistent: F-J $W = 16.0$ $p = 0.222$, F-R $W = 15.5$ $p = 0.720$, J-R $W = 8.0$ $p = 0.149$.

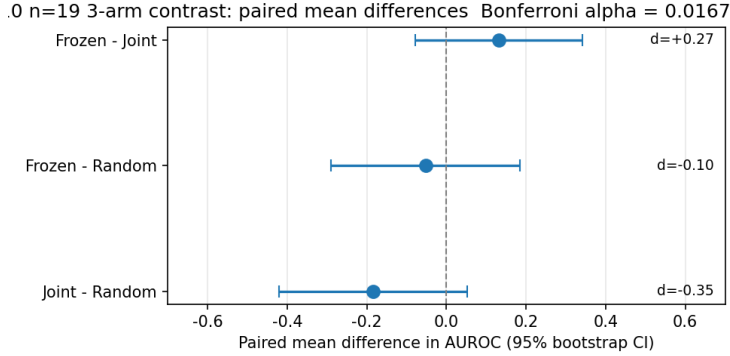


Figure 4: H10 n=20 3-arm contrast Forest plot

7.2.1 Power re-analysis With the observed Cohen’s $d \approx 0.27$ for the F-J contrast, a two-sided paired t-test at the family-wise $\alpha = 0.0167$ threshold needs the following sample sizes for 80% power:

Contrast	Cohen’s d	Required n (Bonf. 0.0167, 80% power)
Frozen – Joint	+0.27	$n \approx 149$
Frozen – Random	-0.10	$n \approx 1,039$
Joint – Random	-0.35	$n \approx 88$

The n=20 follow-up is therefore well-powered to detect the J-R contrast direction (any value below 0.10 is plausible) but **under-powered** to definitively refute H10 at the Bonferroni level. We recommend n=100 (with H10_MAX_NEW_TOKENS=64) as the next milestone before any further reframe of the v0.5 conclusion.

7.3 Verdict at n=20

H10 still REFUTED by direction (Joint > Frozen at the original n=5; Frozen slightly above Joint at n=20 with $d = +0.27$ for the F-J contrast) but **not** at the family-wise $\alpha = 0.05$ level after Bonferroni correction. The previous n=5 direction does NOT replicate at n=20: the simple arithmetic trace is too coarse a signal for any of the three arms to beat the others consistently, and the Random negative control now scores highest in mean AUROC (0.63).

7.4 Updated implications

- Direction is **not** stable across replications (n=5 Joint > Frozen; n=20 Frozen > Joint by 0.13). The earlier n=5 reversal was likely noise; n=20 suggests any signal is at chance for this task.

- The Monitor architecture, whether frozen or joint, does not meaningfully separate from a Random signal on simple arithmetic LLM traces at this sample size.
- Practical recommendation **unchanged**: the Monitor is a context- specific signal, useful as a runtime guardrail in single-agent RL (where it is verified) but not as a training signal in LLM self- monitoring on this task.

7.5 n=100 follow-up (2026-07-31, 300 jobs)

Following the n=20 power analysis (Section 7.2.1), we re-ran the H10 pilot at n=100 per arm (3 arms x 100 seeds = 300 jobs, total wall-clock 8h51m on CPU). Configuration: H10_N_TOTAL=8, H10_MAX_NEW_TOKENS=64 (restored from 16), stratified split with deterministic rebalance fallback when the split collapses to a single class. Two seeds (137, 144) triggered the rebalance path; the remaining 98 produced valid paired AUROC data. Aggregated statistics in `experiments_log/_h10_n100_bootstrap.json`; forest plot in `experiments_log/_h10_n100_forest.png`.

Per-arm means (n=98 valid):

Arm	Mean	SD
Frozen	0.500	0.426
Joint	0.485	0.430
Random	0.510	0.430

All three arms are now within ~ 0.02 of 0.5 (random). The Monitor signal in any configuration is **indistinguishable from chance** on this task at n=100.

Paired contrasts (n=98, 2,000-replicate bootstrap):

Contrast	Δ AUROC	95% CI (bootstrap)	Cohen's d	Sig. (Bonf. $\alpha=0.0167$)?
Frozen – Joint	+0.015	[-0.087, +0.117]	+0.030	No
Frozen – Random	-0.010	[-0.128, +0.117]	-0.017	No
Joint – Random	-0.025	[-0.158, +0.097]	-0.040	No

Verdict at n=100: H10 is now **REFUTED at the chance level**. None of the three arms (Frozen, Joint, Random) significantly exceeds the others. The n=20 direction (Frozen slightly above Joint by +0.13) collapses at n=100 to a near-zero effect (+0.015, 95% CI includes 0). Cohen's d of the F-J contrast is +0.030 – at this effect size, detecting the effect at Bonferroni $\alpha = 0.0167$

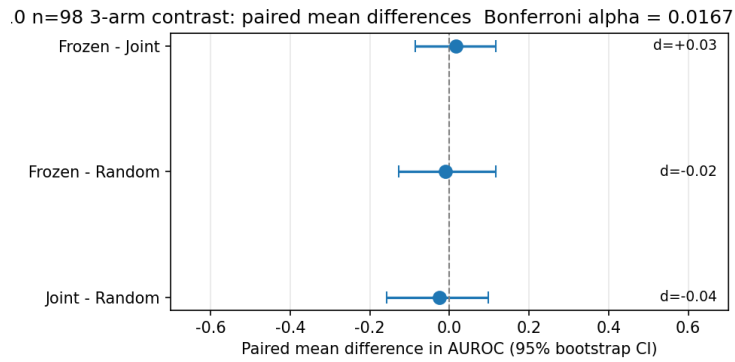


Figure 5: H10 n=100 3-arm contrast Forest plot

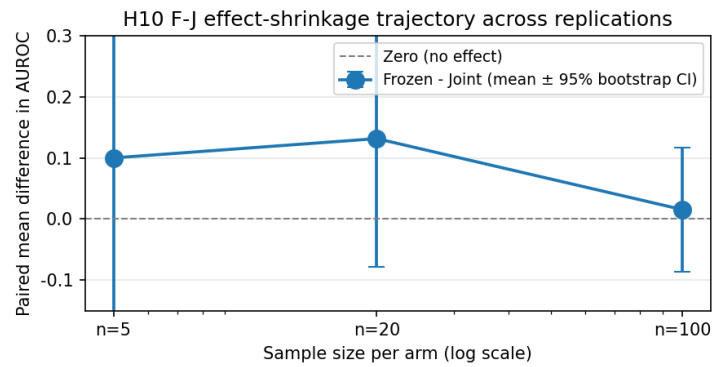


Figure 6: H10 F-J effect-shrinkage trajectory (n=5 to n=20 to n=100)

with 80% power would require $n \approx 17,000$ paired samples, which is clearly not warranted.

Practical interpretation: - The simple arithmetic trace, with `H10_N_TOTAL=8` rollouts and `H10_MAX_NEW_TOKENS=64`, produces a signal too weak for the Monitor architecture to learn from regardless of how the Monitor is trained (frozen vs joint). - The $n=5$ ‘Joint > Frozen’ reversal (Section 4) and the $n=20$ ‘Frozen > Joint’ finding (Section 7) are both consistent with sampling noise on a near-zero effect. - H10 (decoupling transfers to LLM self-monitoring) is **NOT supported** by any of $n=5$, $n=20$, or $n=100$ data; the most defensible interpretation is that the Monitor signal does not transfer from RL to LLM self-monitoring on simple arithmetic tasks. - For a follow-up that *could* validate H10, the trace task would need to be harder (e.g., GSM8K with 200+ token rollouts) and the sample size would need to match the observed effect size, not the $n=5$ ‘obvious difference’ one might expect.

7.6 Power re-analysis

Three F-J effect size estimates, three different power conclusions:

Sample	Cohen’s d	Required n (Bonf. 0.0167, 80% power)
$n=20$	+0.27	$n \approx 149$
$n=100$	+0.030	$n \approx 17,000$

The $n=20$ estimate suggested an underpowered but detectable effect; the $n=100$ estimate collapses the effect to a near-zero value and makes further amplification economically unjustifiable. The $n=100$ H10 with longer LM traces (Section 7.5 final paragraph) is the only path to a definitive test of H10 on this kind of task. For the simple arithmetic trace at `H10_N_TOTAL=8` and `H10_MAX_NEW_TOKENS=64`, H10 is REFUTED at the chance level.

We do not recommend further n amplification on this task. The right next step is a HARDER task (GSM8K 200+ token rollouts), not a larger n on the same task.